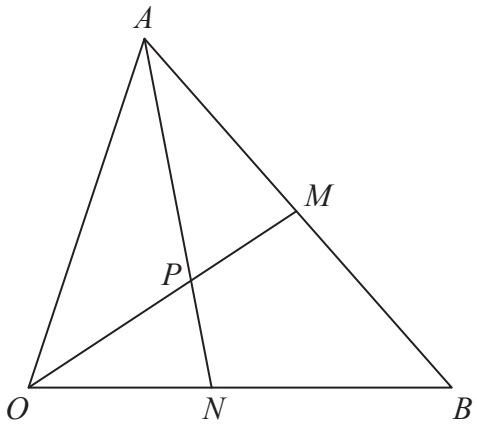
21



OAB is a triangle.
OPM and APN are straight lines.
M is the midpoint of AB.

$\overrightarrow{OA} = \mathbf{a}$ $\overrightarrow{OB} = \mathbf{b}$

$OP : PM = 3 : 2$

Work out the ratio $ON : NB$

$AB = b - a$

1 mark
Vector from A to B

$OM = a + \frac{1}{2}(b - a) = \frac{1}{2}(a + b)$

1 mark
M is the midpoint of AB

$OP = \frac{3}{5}\ OM = \frac{3}{10}(a + b)$

$AP = OP - OA = \frac{-7a + 3b}{10}$

1 mark
Use the 3 : 2 split, then subtract OA

Let $ON = kb$. Then $AN = -a + kb$.

A, P, N are collinear: $AN = t\ AP$.

Compare a: $-1 = \frac{-7t}{10}$ so $t = \frac{10}{7}$

1 mark
Match vector coefficients to find k

Compare b: $k = \frac{3}{10} \times \frac{10}{7} = \frac{3}{7}$

$ON : NB = 3 : 4$

1 mark
N is three sevenths along OB

3 : 4

(Total for Question 21 is 5 marks)